Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The time taken for 90% of a first order reaction to complete is approximately

(A) 1.1 times that of half-life

(B) 2.2 times that of half-life

(C) 3.3 times that of half-life

(D) 4.4 times that of half-life

Q.2

Which of the following reactions is not of the first order?

(A) Inversion of sucrose in the presence of acid

(B) Acid-catalyzed hydrolysis of ethyl acetate

(C) Hydrolysis of tertiary butyl halide using alkali

(D) Oxidation of I^{-} ion by $\mathrm{S}_2\mathrm{O}_8^{2-}$ ion

Q.3

A first order reaction is 50% completed in 20 minutes at $27^{\circ}\mathrm{C}$ and in 5 minutes at $47^{\circ}\mathrm{C}$. The energy of activation of the reaction is :

(A) $43.85\ \mathrm{kJ/mol}$

(B) $55.14\ \mathrm{kJ/mol}$

(C) $11.97\ \mathrm{kJ/mol}$

(D) $6.65\ \mathrm{kJ/mol}$

Q.4

In a first-order reaction $\mathrm{A} \rightarrow \mathrm{B}$, if k is rate constant and initial concentration of the reactant A is $0.5\mathrm{M}$, then the halflife is

(A) $\frac{\log 2}{\mathrm{k}}$

(B) $\frac{\log 2}{\mathrm{k}\sqrt{0.5}}$

(C) $\frac{\ln 2}{k}$

(D) $\frac{0.693}{0.5\mathrm{k}}$

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Q.5

A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial velocity is ten times the permissible value, after how many days will it be safe to enter the room?

(A) 100 days

(B) 1000 days

(C) 300 days

(D) 10 days

Q.6

Half life period of a first-order reaction is 1386 seconds. The specific rate constant of the reaction is:

(A) $0.5 \times 10^{-2}\ \mathrm{s}^{-1}$

(B) $0.5 \times 10^{-3}\ \mathrm{s}^{-1}$

(C) $5.0 \times 10^{-2}\ \mathrm{s}^{-1}$

(D) $5.0 \times 10^{-3}\ \mathrm{s}^{-1}$

Q.7

$\mathrm{t}_{1/4}$ can be taken as the time taken for the concentration of a reactant to drop to $\frac{3}{4}$ of its initial value. If the rate constant for a first order reaction is k, the $t_{1/4}$ can be written as

(A) $0.75/k$

(B) $0.69/k$

(C) $0.29/k$

(D) $0.10/k$

Q.8

$3\ \mathrm{A} \rightarrow \mathrm{B} + \mathrm{C}$, it would be a zero order reaction when

(A) the rate of reaction is proportional to square of concentration of A

(B) the rate of reaction remains same at any concentration of A

(C) the rate remains unchanged at any concentration of B and C

(D) the rate of reaction doubles if concentration of B is increased to double

Q.9

For the first order reaction
$\mathrm{C_2H_4O(g)} \rightarrow \mathrm{CH_4(\ g)} + \mathrm{CO(g)}$, the initial pressure of

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$C_2H_4O(g)$ is 80 torr and total pressure at the end of 20 minutes is 120 torr. Calculate the time in minutes needed for 75% decomposition of C_2H_4O.

(A) 30

(B) 38

(C) 42

(D) 40

Q.10

The half life period for catalytic decomposition of AB_3 at $50\ mmHg$ is $4hrs$ and at $100\ mmHg$ it is $2hrs$. Find the order of reaction.

(A) 2

(B) 4

(C) 3

(D) 1

Q.11

The decomposition of a substance follows first order kinetics. Its concentration is reduced to $1/8$ th of its initial value in 24 minutes. The rate constant of the decomposition process in min^{-1} is

(A) 0.087

(B) 0.085

(C) 0.87

(D) 0.85

Q.12

The rate law for a reaction between the substances A and B is given by rate $= k[A]^n[B]^m$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as

(A) $(m+n)$

(B) $2^{(n+m)}$

(C) $2^{(n-m)}$

(D) $\frac{1}{2^{(m+n)}}$

Q.13

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The halflife of a radioactive element is 20 min. The time interval between the stages of its 33% and 67% decay is

(A) 40 min

(B) 20 min

(C) 30 min

(D) 25 min

Q.14

The reaction $\mathrm{A} \rightarrow \mathrm{B}$ follows first order kinetics. The time taken for 0.8 mole of A to produce 0.6 mole of B is 1 hour. What is the time taken for conversion of 0.9 mole of A to produce 0.675 mole of B?

(A) 2 hours

(B) 1 hour

(C) 0.5 hour

(D) 0.25 hour

Q.15

The dependence of the rate constant for a reaction on temperature is given by the equation $k = Ae^{-E_a / RT}$. Under what conditions is the rate constant k the smallest?

(A) High T and large $\mathrm{E_a}$

(B) High T and small $\mathrm{E_a}$

(C) Low T and large $\mathrm{E_a}$

(D) Low T and small $\mathrm{E_a}$

Answers & Solutions

Q.1 Answer:

3.3 times that of half-life

Solution:

$t_{90\%} = \frac{2.303}{k} \log \frac{100}{100-90}$ **(I)**
$t_{50\%} = \frac{2.303}{k} \log \frac{100}{100-50}$ **(II)**
Dividing $\frac{t_{90\%}}{t_{50\%}} = \frac{\log 10}{\log 2}$
$\therefore t_{90\%} = 3.3 t_{50\%}$

Q.2 Answer:

Oxidation of I^- ion by $\mathrm{S_2O_8^{2-}}$ ion

Solution:

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It is a second order reaction, first order both w.r.t $S_2O_8^{2-}$ and I^-.

$\therefore r = k\left[S_2O_8^{2-}\right][I^-]$

All other options are of first order reaction.

Q.3 Answer:

$55.14\ \text{kJ/mol}$

Solution:

$k_{1(300)} = \frac{0.693}{20}; k_{2(320)} = \frac{0.693}{5}$

$\text{In}\frac{k_2(320)}{k_{1(300)}} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

$E_a = \frac{2.303RT_1\,T_2}{(T_2 - T_1)}\log\frac{k_2}{k_1}$

$= \frac{2.303\times 8.314}{20\times 1000}\times 300\times 320\log 4$

$= 55.14\ \text{kJ/mol}$

Q.4 Answer:

$\frac{\ln 2}{k}$

Solution:

For a first order reaction

$k = \frac{2.303}{t}\log_{10}\frac{a}{a-x}$

when $t = t_{1/2}$

$k = \frac{2.303}{t_{1/2}}\log_{10}\frac{a}{a-a/2}$

or $t_{1/2} = \frac{2.303}{k}\log_{10}2 = \frac{\ln 2}{k}$

Q.5 Answer:

100 days

Solution:

Since initial velocity is ten times the permissible value $\therefore A_0 = 10\ A$

$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{30}$

$t_{1/2} = \frac{2.303}{\lambda}\log\frac{A_0}{A} = \frac{2.303}{0.693/30}\log\frac{10\ A}{A}$

$= \frac{2.303\times 30}{0.693}\times\log 10 = 100$ **days.**

Q.6 Answer:

$0.5\times 10^{-3}\ s^{-1}$

Solution:

For a first order reaction

$t_{1/2} = \frac{0.693}{k}; k = \frac{0.693}{1386} = 0.5\times 10^{-3}\ s^{-1}$

Q.7 Answer:

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$0.29/k$

Solution:

$$t_{1/4} = \frac{2.303}{k}\log\frac{1}{3/4} = \frac{2.303}{k}\log\frac{4}{3}$$
$$= \frac{2.303}{k}(\log 4 - \log 3) = \frac{2.303}{k}(2\log 2 - \log 3)$$
$$= \frac{2.303}{k}(2 \times 0.301 - 0.4771) = \frac{0.29}{k}$$

Q.8 Answer:

the rate of reaction remains same at any concentration of A

Solution:

For reaction $3\ \mathrm{A} \longrightarrow \mathrm{B} + \mathrm{C}$
If it is zero order reaction $\mathrm{r} = \mathrm{k}[\mathrm{A}]^0$**, i. e the rate remains same at any concentration of '** A'**. i.e independent upon concentration of A.**

Q.9 Answer:

40

Solution:

Let x **torr of** $\mathrm{C_2H_4O}$ **decompose after** $20\ \mathrm{min}$**. Then,** $80 - x + 2x = 120; x = 40$ **torr** $= 50\%$ **of initial pressure.**
Hence $t_{1/2} = 20\ \mathrm{min}$**. For** 75% **reaction, fraction left**
$= \frac{25}{100} = \frac{1}{4} = \left(\frac{1}{2}\right)^2$
No. of half lives $= 2$**. Time needed for** 75% **reaction.,** $2 \times 20 = 40\ \mathrm{min}$

Q.10 Answer:

2

Solution:

$\mathrm{t_{1/2}} \propto \frac{1}{(\mathrm{p})^{\mathrm{n}-1}}$**, where** n **is the order of reaction**
$\frac{2}{4} = \left(\frac{50}{100}\right)^{\mathrm{n}-1}$ **or** $\frac{1}{2} = \left(\frac{1}{2}\right)^{\mathrm{n}-1}$
$\mathrm{n} = 2$

Q.11 Answer:

0.087

Solution:

$\mathrm{k} = \frac{2.303}{\mathrm{t}}\log\frac{\mathrm{a}}{\mathrm{a}-\mathrm{x}} = \frac{2.303}{24}\log\frac{1}{\frac{1}{8}} = \frac{2.303}{24}\log 8$
$= 0.087\ \mathrm{min}^{-1}$

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Q.12 Answer:

$2^{(n-m)}$

Solution:

Rate $_1 = k[\ \mathrm{A}]^n[\ \mathrm{B}]^m$; **Rate** $_2 = k[2\ \mathrm{A}]^n[1/2\ \mathrm{B}]^m$

$\therefore \frac{\text{Rate}_2}{\text{Rate}_1} = \frac{k[2\ \mathrm{A}]^{\mathrm{n}}[1/2\ \mathrm{B}]^{\mathrm{m}}}{k[\ \mathrm{A}]^{\mathrm{n}}[\mathrm{B}]^{\mathrm{m}}}$

$= [2]^{\mathrm{n}}[1/2]^{\mathrm{m}} = 2^{\mathrm{n}} \cdot 2^{-\mathrm{m}}$

$= 2^{\mathrm{n}-\mathrm{m}}$

Q.13 Answer:

$20\ \mathrm{min}$

Solution:

Change in 67% **to** 33% **is almost half the concentration change.**

$67\% \xrightarrow[\text{change}]{\text{Half}} 33.5 (\approx 33\%)$

So time interval between the stages of its 33% **and** 67% **decay is same as** $t_{1/2} = 20\ \mathrm{min}.$

Q.14 Answer:

1 hour

Solution:

$\mathrm{A} \rightarrow \mathrm{B}$ **For a first order reaction**

Given $a = 0.8\ \mathrm{mol}, (a - x) = 0.8 - 0.6 = 0.2$

$k = \frac{2.303}{1} \log \frac{0.8}{0.2}$ **or** $k = 2.303 \log 4$

again $a = 0.9, a - x = 0.9 - 0.675 = 0.225$

$k = \frac{2.303}{t} \log \frac{0.9}{0.225}$

$2.303 \log 4 = \frac{2.303}{t} \log 4$

Hence $t = 1$ **hour**

Q.15 Answer:

Low T and large E_a

Solution:

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